

Intelligent Factory Agents with Predictive Analytics

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1.1 Introduction / Motivation	1
1.2 Application Overview	3
1.3 Application Details	5
1.4 Benefits and Assessment	9
1.5 Discussion	10
1.6 Conclusions	11
References	11

1.1 Introduction / Motivation

Many advanced countries, whose economic base is manufacturing industry, made efforts to improve their up time and production quality because they have more critical challenges from emerging markets and global manufacturing supply chain. Manufacturing firms not only seek manufacturing technique innovation but also began to focus on how to transform their factory based on existing information communication technologies. As a result, technological innovations have been drivers of the evolution of manufacturing paradigms from mass production, through the concepts of lean, flexible, reconfigurable manufacturing, to the current stage of predictive manufacturing characterized by bringing transparency to manufacturing assets capabilities. With this manufacturing transparency, management then has the right information to determine facility-wide OEE. Beyond that, the revealed manufacturing data can be analyzed and transformed into meaningful information to enable prediction and prevention. With the prediction capability, factory asset can be managed cost effectively with just-in-time maintenance.

In regard to the above-mentioned trend, Industry 4.0 is now a new buzzword in manufacturing industry. Under the concept of Industry 4.0, intelligent analytics and cyber-physical systems (mentioned in [1]) are teaming together to realize a new thinking of production management and factory transformation. Table 1 compares the difference between today's factory and an Industry 4.0 factory. In current manufacturing environment, there might be different data sources including sensors, controllers, networked manufacturing systems, etc. In today's factory, component precision and machine throughput is key to success. How to utilize data to understand current condition and detect faults are important research topics. [2-6] For production systems, many commercialized manufacturing systems are deployed in order to aid shop managers in acquire OEE (Overall Equipment Effectiveness) information. Compared with an Industry 4.0 factory, instead of only fault detecting or condition monitoring, components will also be able to achieve self-aware and self-predict capabilities. That is, the health degradation and remaining useful life will be revealed so that more insight will be brought to factory users.

Beyond that, machines health can be predicted based on a fusion of component conditions and peer-to-peer comparison. With the prediction capability, machines can be managed cost effectively with just-in-time maintenance, which eventually optimize machine up time. Finally, historical health information can be fed back to the machine or equipment designer for closed loop lifecycle redesign, and users can have a worry-free productivity.

		Today's Factory		Industry 4.0 Factory	
	Data source	Key attributes	Key technologies	Key attributes	Key technologies
Component	Sensor	Precision	Smart Sensors and Fault Detection	Self-Aware Self-Predict	Degradation Monitoring & Remaining Useful Life Prediction
Machine	Controller	Producibility & Performance (Quality and throughput)	Condition-based Monitoring & Diagnostics	Self-Aware Self-Predict Self-Compare	Up Time with Predictive Health Monitoring
Production System	Networked manufacturing system	Productivity & OEE	Lean Operations: Work and Waste Reduction	Self-Configure Self-Maintain Self-Organize	Worry-free Productivity

Table 1: Comparison of today's factory and an Industry 4.0 factory

The predictive manufacturing combines the information from the manufacturing system and supply chain system. Traditionally, the manufacturers make decision by using the supply chain system, which optimizes the costs by leveraging the logistics and synchronizing supply with demand and measuring performance globally [7]. However, the costs spent on the maintenance and its problems arouse great costs on the asset management. Thus, **predictive manufacturing consists with two parts, supply chain management and predictive maintenance.** On one hand, the smart supply chain management gives key performance indicators by analyzing the historical data including the supplier source, financial data, market consumption, and it predicts and quantifies the leading indicators based on all the read drivers of the business [8-10]. This does not consider the effects of unpredicted downtime and maintenance of the operational performance. On the other hand, the predictive maintenance detects the greatest risk based on gathering real time information like maintenance logs, performance logs, monitoring data, inspection reports, and environmental data, etc. [11-13] Once the risk from certain parts reaches the threshold level, a proactive maintenance will be performed in order to prevent the down time. According to the risk analysis, the production line can only schedule pre-maintenance before the failure happens which can greatly reduce the high cost of fixed schedule maintenance. In addition, it is easy to anticipate the potential problems when customers use the products, which can improve the warranty service and reduce its costs. The

predictive maintenance methodologies are consisted with data information transformation, prediction, optimization and synchronization [13]. Because the maintenance plays an important part in the asset management process [14], the appropriate application of predictive maintenance greatly reduces the costs spending on the unexpected operation problems, and meanwhile, it can provide proper information to the supply chain management like rescheduling the order placements, inventory management, adjusted warranty services, etc., in order to take proactive movements to prevent causing interruption for the supply chain system.

In this chapter, we focus on how to realize predictive manufacturing through the deployment of intelligent factory agent and implementation of analytic tools. The former is in charge of the data flow based on a 5S systematic approach consisting of Sensing, Storage, Synchronization, Synthesis and Service, while the latter is an important key to information transformation using four sub tools: signal processing and feature extraction, health assessment, performance prediction, and fault diagnosis.

1.2 Application Overview

To enable an intelligent factory agent, a system including three subsystems, which are *real-time machine condition monitoring subsystem*, *predictive analytic subsystem*, and *service dashboard subsystem*, is proposed based on current information communication technology. The framework of this system is shown in Figure 1. First of all, the controller signals and sensor data can be extracted automatically from online process by real-time machine condition monitoring subsystem, which can reveal the condition of machine operation, the NC program of the machine, and the corresponding control parameters in real time. Secondly, by analyzing these signals, a set of predictive tools is utilized to assess the health condition, production quality and stability of the current process, in order to achieve a smart asset management. The asset information, which is learned by online PHM modules, will be mapped with original distributions generated by sensor signals, control parameters, e-manufacturing system data and maintenance history. The information will be stored, sorted, and streamlined according to its work order identification by the predictive analytic subsystem. Finally, the factory managers and manufacturing operators can make query and retrieve related analyzed information for optimization of the manufacturing process through the service dashboard provided.

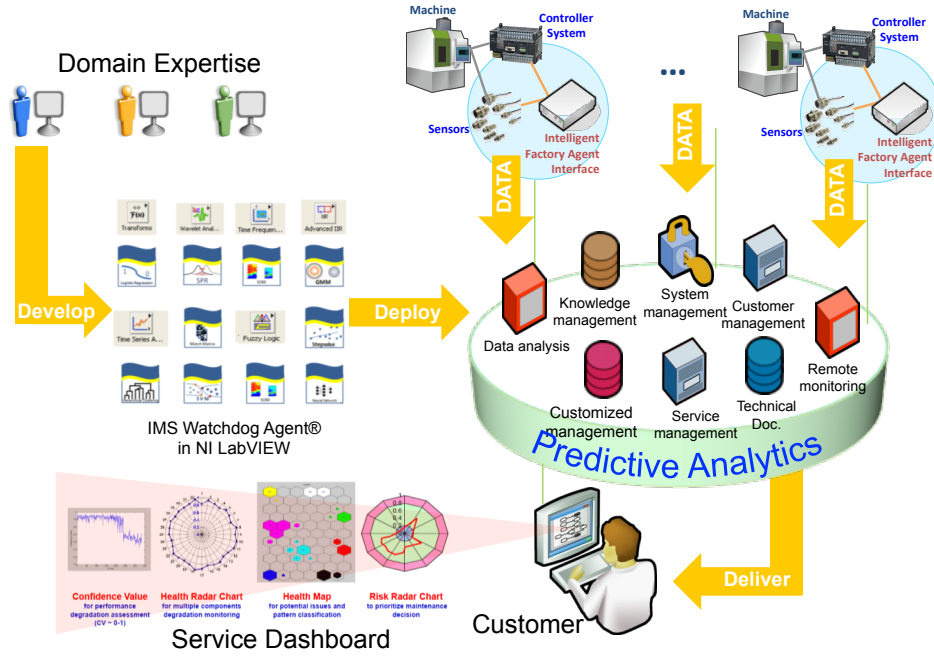


Figure 1: Framework for Intelligent Factory Agents with Predictive Analytics

To acquire the operation condition from machines, frameworks for extracting controller signal from CNC systems has been designed and demonstrated in previous research [15]. In a *real-time machine condition monitoring subsystem*, a controller signal extractor and real-time monitoring module are integrated in an industrial PC. Each machine is equipped with a controller and RJ-45 Internet cable is used to connect between controller and the signal extractor through the Ethernet Internet protocol. It includes a parameter extractor (connect with controllers and extract parameters), rule configuration module (write rules and trigger points for parameter extraction), command receiver (receive the commands from real-time monitoring module and configure accordingly), and an information sender (transfer extracted signals to real-time monitoring module).

Since the sampling frequency of the controller signal extractor is high, the communication between the controller signal extractor and the real-time monitoring module is frequent. To efficiently process the data set and at the same time provide a real-time, stable data extraction performance, an asynchronous transmission mechanism is designed and adopted. Once the controller signal extractor extract the signal data, the data set will be sent to a message queue instead of monitoring module directly. The message queue is a high performance buffer space and can be easily customized to receive and store the data. Once the data set is reserved in the message queue, the real-time monitoring module will use its Listener to register to the message queue. When there are messages waiting in the message queue, the Listener will be triggered to extract the messages from the queue and parse them. The parsed information

will then be sent to real-time monitoring module to analyze and become visualized information through user interface.

To achieve real-time predictive manufacturing goals, a platform for increase productivity and advanced manufacturing with analytic toolset should be designed and implemented. When there are more and more machine tools in a factory, a *predictive analytic subsystem* can grasp the machine tools' real-time condition, monitoring operation and detecting production quality and performance. The subsystem leverages the machining information extracted from machine tools and collect productivity information from production lines. It eventually forms a machine service network to help customers manage their production line in real-time and more intelligent. While the domain-specific knowledge of machining process is accumulating, it can also support factory users to find out the weakness in the current manufacturing process, which means the quality of their manufacturing can also be increased. Besides, the knowledge collected can become an expert or reference and be provided to customers. Before the abnormal condition occurs, take the actions to avoid the possible damages and stabilize the production quality level. Finally, factory users can manage the overall productivity and future efficiency of their equipment through *real-time service dashboard subsystem*, since it offers factory managers the flexibility and knowledge toward their equipment and factory based on a set of designed visualization modules.

1.3 Application Details

To illustrate the idea of the factory industrial agent previously mentioned, in this section several applications are introduced from different points of view.

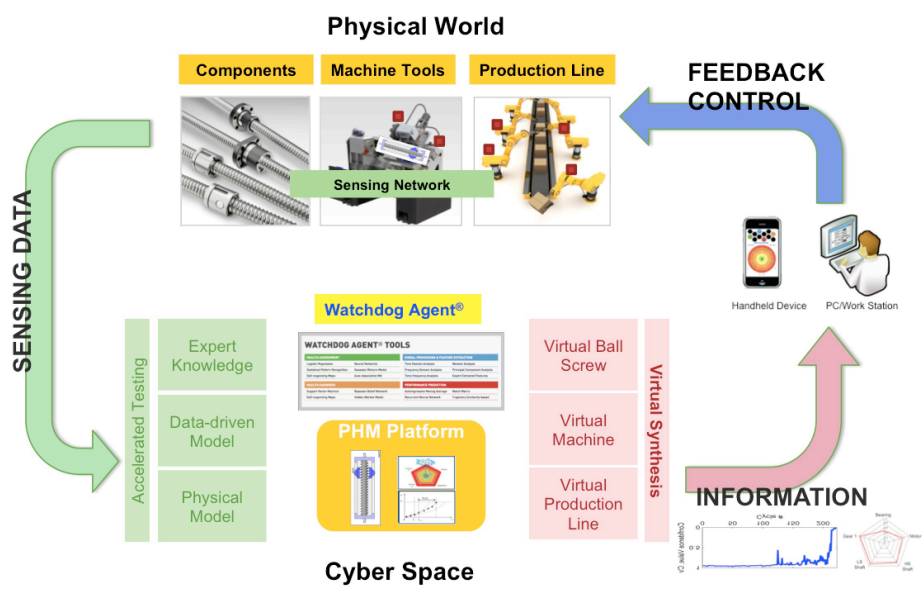


Figure 2: An example of industrial agent framework for components, machines, and production lines

Smart factory transformation from component level point of view

The increasing feeding rate and high acceleration value have improved the machine tool efficiency and accuracy and also reduced the machining times. Each component of machine tools can directly affect the overall precision of the machine tool system. The critical components can be defined based on their criticality level, low failure frequency but high impact. The condition-based monitoring can enable the manufacturing factories to observe the degradation of each critical component and calculate their the health value, therefore the overall health condition of the machine will be evaluated based on the health condition of each component. This health indicator called the confidence value can provide and predict the information of the machine's health condition. Hence, the predictive analytical algorithms provide significant improvements of adding PHM to traditional maintenance schemes. Machine data is effectively transformed by PHM algorithms into valuable information that can be used by factory managers to optimize production planning, save maintenance cost and minimize equipment downtime.

In a smart factory, all the physical entities in the factory are connected together physically and virtually. In the physical space, the sensors are utilized to measure and detect the occurrence of a fault or anomaly, and isolate the location of the fault and identify the failure types. Beside the sensor signals, the outputs from the machine controller like current, voltage, power consumption can also be used to reflect the machines' overall health degradation. These physical entities are connected in the cyber space through the Internet. The sensor signals are collected through data acquisition hardware, and either stored locally or uploaded to a cloud server. After storing the data, pre-processing techniques are employed to check the data quality and increase the signals' signal-to-noise ratio (SNR), such as the data filtering, quantization, etc. Data segmentation is used to select the interesting segment from the input signals. Furthermore, the features (health indicators) are extracted, and a health value is calculated over time to evaluate the behavior of the machine components and detect incipient signs of failure even before the actual failure event. In addition to this data-driven method, the physical model and experimental knowledge are also applied to fully understand the performance of these components, like the finite element analysis (FEA) models, modal analysis, etc. In the cyber space, all of these models are integrated to convert the data to information, with which the customers can schedule the timely maintenance before the failure.

In order to model the components' behavior quickly and effectively, the accelerated degradation test in the lab is utilized in the machine reliability analysis, the different stress factors are screened and the important stress factors are picked according to the quantization matrix. After that, the characteristics tests are conducted to identify the initial condition. Thus the data

collected from multiple degradation process under different stress levels can generate the degradation pattern for the component, thus the time conversion between these different degradation pattern will be validated, which can assist to identify the degradation trend in the actual industrial application. After establishing the degradation model, this PHM model will be integrated in the cyber space. The data collected from the machine components will be processed through this PHM model and thus a virtual component is modeled in the cyber space, so it can simulate and predict the health pattern of the machine component in the reality.

The component level of the machine system includes the critical components like motors, bearings, ball screws, gearbox, etc. These critical components determine the performance of the machine system. Thus, **solutions provided for the cyber-physical system of component levels will be:**

1. Data acquisition, sensor installation in physical space
2. Establish the virtual model for the critical components using PHM solutions in cyber space
 - a. Component mechanism analysis and failure detection, for example:
 - bearing inner race/outer race failure and ball wear, etc.;
 - ball screw's preload loss, internal surface wear, etc.;
 - gearbox's broken teeth, bent shaft, etc.;
 - b. Prognostics and Remaining Useful Life prediction for the components.
 - Use the prediction model to estimate the life of each component, and evaluate the life of whole system
 - c. Update the virtual model using the new data from the field operation data

Once the condition of machine components being analyzed and predicted, machinery users and component suppliers can work together to improve up time and reduce cost by an optimal management strategy. Of course, it won't happen without component transparency and remaining useful life prediction.

Smart factory transformation from machine level point of view

Machine monitoring systems put more focuses on the data acquisition and storage, but there is less emphasis on the analysis of the data; the data contains valuable health information that can be used to support factory decision-making procedures. Machinery prognostics and health management methodologies are able to make the whole process more systemized.

The traditional approach of reactive maintenance - essentially repairing a machine when it fails - may seem like the simplest option, but this approach is clearly inadequate in a modern factory. As throughput times have become increasingly fast due to improvements in plant automation, unexpected breakdowns have become prohibitively expensive and even catastrophic. While more recent preventive maintenance strategies [16-19] may offer higher availability through time-based conditioning/repair/replacement activities that

preclude unexpected downtime, this approach also has two major disadvantages. Firstly, preventive maintenance is an expensive program to maintain, especially if its intervals are kept very tight. Secondly, although preventive maintenance activities ensure that components do not fail or exhibit significant behavioral changes, there is no insight learned about the equipment's actual degradation cycle that can be used to improve its design.

Therefore, beyond what we have discussed on component level analytics, predictive analytics should be extended to a whole machine level. By trending degradation patterns, it can predict, with some level of confidence, when equipment is going to reach a failure condition. With the use of advanced predictive tools and algorithms, manufacturing asset behavior is modeled and tracked using a set of metrics known as “health value” or “confidence value.” Finally, prediction tools are utilized to infer when the machine is likely to fail. With such information, a much higher level of manufacturing transparency is achieved. Maintenance and production personnel can then collaboratively and proactively plan when to schedule repair/conditioning activities to avoid equipment failure so it does not interfere with planned production goals.

Besides, from the control perspective, there are two basic categories of machine errors, quasi-static errors and dynamic errors. Machine error compensation does not aim to reduce the absolute value of errors, but the effects of these errors on the machining accuracy and final dimensions of produced parts. The degree to which machining accuracy can be achieved by error compensation is highly dependent on the repeatability of the machine itself and the methods selected to demonstrate the interconnection between different errors. Currently, machine compensations, especially, quasi-state errors is achieved through off-line error calibration and modeling, and on-line error correction. This method is highly dependent on the accuracy of the preset calibration table, thus it is labor-intensive and time-consuming. An on-line error measurement and compensation would alleviate the issues associated with the traditional method. Optical error measurement system installed on the machine would be able to measure the machine errors on-line and feed these errors back to the machine control module, which would then calculate the compensated values and make on-line error correction. This closed-loop method would reduce the workload for the machine error calibration and significantly improve the machining accuracy.

Smart factory transformation from factory level point of view

A factory or manufacturing facility can be described as a 5M system [1], which consists of Materials (properties and functions), Machines (precision and capabilities), Methods (efficiency and productivity), Measurements (sensing and improvement), and Modeling (prediction, optimization, and prevention). To realize a smart factory, the “Big Data” coming from the 5M systems should be further analyzed and integrated in a systematic way.

Big data has raised opportunities to develop “smarter” decision support tools in the factory. Especially, the increasing amount and accuracy of the real-time data from the Factory Information System (FIS) can be utilized to make more effective and dynamic on-line operational policies. For example, how to change the production sequences if there are some demand changes on the market? What is the bottleneck machine of the system in real-time? Which maintenance tasks should be conducted at when and by whom, so that the overall cost in the system can be minimized? What is the opportunity time window to stop one machine for maintenance while still satisfying the system throughout requirement? Analyzing the real-time system condition and predicting its evolution can aid factory managers in answering these questions.

For example, by making the manufacturing capability transparent, plant and corporate managers have the right information to assess facility wide overall equipment effectiveness (OEE). Also, with the use of such advanced prediction tools, companies can plan more cost-effective, just-in-time maintenance to ensure equipment health over a longer period.

Also noted that, to make optimal decisions, we need the information integrated from component level, machine level and the system levels. For example, to make an optimal sequence of maintenance task, we need to know: the current healthy state of the machines and how they will degrade over time, the availability and skill levels of the maintenance crews, the inventory level, and also the production requirement of the system, etc. The more accurate this information is, the more effective these decision support tools will be.

1.4 Benefits and Assessment

Predictive manufacturing is to apply predictive analytical methods for different aspects during the manufacturing process. Respectively, with intelligent factory agents companies will be able to achieve the following benefits:

1. Schedule in-time maintenance and prevent unexpected machine downtime: up time of manufacturing line is directly related to the profits and for some industries even a short period of product line failure would cost much financially. A properly developed and implemented predictive maintenance scheme can reduce the risk of unscheduled downtime and save costs.
2. For product quality – first part correct: predict the quality of the product even before it is manufactured. The degradation of machines does not only increase the risk of potential downtime but also affect the quality of the parts being produced. As a consequence, a predictive strategy can be used to suggest and schedule maintenance actions for key components such as machine tools even before the actual failure occurs to maintain the product quality.
3. For smarter logistics and supply chain management: based on the predicted customer needs and machine condition, the inventory,

budgeting and purchasing actions can be made accordingly to save cost.

The designed system, which can also be integrated with the cloud-computing paradigm, brings the health monitoring systems to a new level. Besides the benefits of convenience in connectivity and management, and real-time computing, this system can have the following superiorities compared to other monitoring systems:

1. Accurate and generic machine health assessment and prognostics – Since various PHM techniques are in-clouded in the APP pool as modules, they can easily be combined and implemented to generate customized workflows. Due to this standardization, different combinations of tools can be tested to assure the accuracy of the output.
2. Rapid service deployment – The configuration file mentioned before can be constructed based on the expert knowledge or previous experiences for similar cases. This makes the new workflows to be easily developed and make the deployment services fast and convenient.
3. High level of customization – The APP pool, as mentioned before, can provide the user with various options. As these options are easy for the user to try, one can customize their own framework for the best results.
4. Knowledge integration for data mining purposes – As the obtained data can be stored during time and classified based on sources, working regimes etc., the data mining and knowledge discovery tasks will be facilitated for further improvement of PHM field.
5. Easy-to-update – All the instances (virtual machines) of an APP module are eventually images, which can be updated once an instance is initialized. This system update can be very beneficial for the manufacturing units with a fleet of equipment.

1.5 Discussion

The implementation of intelligent factory agent and predictive analytic methods in manufacturing can be called predictive manufacturing process control. Predictive manufacturing process control provides real-time report on the trend of manufacturing quality that enables immediate corrective action. If the critical information from the manufacturing line is processed in the post-production environment, there will be no option for having immediate corrective actions to avoid the reduction of the manufacturing capabilities of the production lines. For example, in Porsche manufacturing plant, the shape of the car bodies and their precise positioning in front of the robots are measured instantly by a measuring system. This system measures the dimensions and position of each body instantly with a non-contacting method. This measuring system helps the quality of the production to be at the same level since any

minor deviations from the normal operation will be detected and predictive / corrective actions performed [20].

Predictive Manufacturing may also be referred as Manufacturing Intelligence Technology. This is the next generation of manufacturing decision-makers, which fuels the need to ensure that minute-by-minute decisions in the control room are aligned with the corporate business strategy. To put in other words, it helps to bridge the gap between production and business environments. On the other hands, model-based predictive manufacturing intelligence solutions have the capability to put forward events that are about to happen, providing manufacturers with the functionality to test scenarios so that operators can make optimal decisions before events occur. Predictive solutions can provide insight into the business impact of operating decisions by using process models [21].

1.6 Conclusions

While information communication technologies become mature and easily adopted, factories are tend to have more data from add-on sensors, machine controllers, metrology, maintenance history, and all kinds of e-manufacturing systems. How to process the data for a deep understanding and insight on factory assets is definitely a key for manufacturing industries to be competitive in this Industry 4th revolution.

In this chapter, a framework for industrial agent for factory asset management is introduced and applications are demonstrated from different points of view, including component level, machine level and factory level. The proposed agent with the capability on predictive analysis can benefits manufacturers to acquire the health and quality information of their factory asset earlier and improve current process immediately. Eventually, it can optimize the manufacturing process and also can extend to the collaboration among manufacturers, suppliers and customers.

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